

Preliminary Biodiversity Net Gain (BNG) Assessment

Project: Proposed Water Infrastructure & Wetland Retention Basin

Site Area: 5.1 h

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1. Introduction & Site Context

This report is based on a proposed 5.1-hectare water-infrastructure project and outlines its preliminary Biodiversity Net Gain (BNG) calculations. The development plan includes a new constructed wetland retention basin alongside operational infrastructure. The primary aim of this assessment is to highlight whether the current design meets the mandatory 10% Biodiversity Net Gain target required for planning permission using the statutory DEFRA metrics, and if not, the recommendations to achieve this target.

2. Baseline Habitats (Pre-Development)

The baseline habitats were mapped using ArcGIS Pro based on UKHab classifications (using satellite context). The site is made up of three habitats: the majority is of low-value modified grassland (agricultural management), connected to moderate-quality neutral grassland and separated by moderate value mixed scrub.

In total, the existing site generates **20.83 Baseline Habitat Units**.

UKHab Code	Habitat Type	Area (ha)	Condition
g4	Modified grassland	3.34	Poor
g3c	Other neutral grassland	1.19	Moderate
h3	Mixed scrub	0.58	Moderate

3. Proposed Design & BNG Outcome

The proposed site replaces just over half the site (2.52 ha) with hard infrastructure (developed sealed surface), which equates to 0 in the metric. In attempt to compensate for the loss, the new design creates a new wetland habitat, extends the mixed scrub and enhances the retained neutral grassland.

Despite these habitat creation efforts, the overall loss created by the new hard infrastructure, results in a negative score. The post-development landscape generates **15.28 Habitat Units**.

UKHab Code	Habitat Type	Area (ha)	Condition
u1b	Developed sealed surface (Lost)	2.52	N/A (0 units)
g3c	Other neutral grassland (Enhanced)	0.93	Good
r1	Constructed wetland (Created)	0.82	Moderate
h3	Mixed scrub (Expanded)	0.83	Moderate

Figures 1 and 2 below show a side-by-side comparison of baseline (left) and proposed (right) habitats created in ArcGIS Pro. Post mapping, the attribute data was exported and processed using a custom automated Python and in-memory SQLite database.

Figure 1 : Baseline UKHab Habitat Plan



Figure 2 : Proposed Habitat Creation & Retention Plan



Metric Summary:

Total Baseline Units: 20.83

Total Proposed Units: 15.28

Net Unit Change: -5.55

Percentage Change: -26.64%

Conclusion: The current design fails the statutory +10% target.

4. Recommendations for Achieving 10% BNG

To pass planning, the project needs to achieve a minimum of **22.91 Habitat Units** (+10% of the baseline). This leaves us with a current shortfall of **7.63 units**.

Below are some options to be discussed with the project management team to close this gap:

-Review the Infrastructure Footprint: Can the engineers reduce the 2.52 ha developed sealed surface, or swap some of the solid concrete to permeable paving?

-Expand the Wetland Area: If there is flexibility in the site layout, we could look at expanding the 0.82 ha constructed wetland.

-Look into Off-Site Mitigation: If the 2.52 ha infrastructure is necessary, we should start looking for local Habitat Banks within our Local Planning Authority Area to buy off-site units.

-Statutory Credits: If none of these options are available, our final option is to buy Statutory Biodiversity Credits from Natural England, however, this is expensive and a last resort.